

Inverting buck-boost converter: Verified State-Space Averaging

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Assumptions and notation

The converter is ideal, operates in continuous-conduction mode at fixed switching frequency, and drives a resistive load R . The state vector is $x = [i_L \ v_o]^T$, D is the ON duty ratio, and $D' = 1 - D$. For the inverting buck-boost, v_o is the physical negative output voltage.

Independent hand derivation

Kirchhoff's laws give

$$ON : L\dot{i}_L = v_g, \quad C\dot{v}_o = -v_o/R, \quad (1)$$

$$OFF : L\dot{i}_L = v_o, \quad C\dot{v}_o = -i_L - v_o/R. \quad (2)$$

These laws were entered independently of the MNA matrices and are used as the verification oracle.

Switching topology models from Rust MNA

$$\dot{x} = A_{on}x + B_{g,on}v_g, \quad A_{on} = \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{CR} \end{bmatrix}, \quad B_{g,on} = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}, \quad (3)$$

$$\dot{x} = A_{off}x + B_{g,off}v_g, \quad A_{off} = \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{CR} \end{bmatrix}, \quad B_{g,off} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (4)$$

Input-to-output state-space representation

State-space averaging gives

$$\dot{\hat{x}} = A\hat{x} + B_g\hat{v}_g, \quad \hat{v}_o = C_y\hat{x}, \quad C_y = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad (5)$$

where

$$A = \begin{bmatrix} 0 & -\frac{D-1}{L} \\ \frac{D-1}{C} & -\frac{1}{CR} \end{bmatrix}, \quad B_g = \begin{bmatrix} \frac{D}{L} \\ 0 \end{bmatrix}. \quad (6)$$

The corresponding transfer function is

$$G_{vg}(s) = \frac{\hat{v}_o(s)}{\hat{v}_g(s)} = C_y(sI - A)^{-1}B_g = \frac{DR(D-1)}{CLRs^2 + D^2R - 2DR + Ls + R}. \quad (7)$$

Control-to-output state-space representation

The steady operating point is

$$X = -A^{-1}B_gV_g = \begin{bmatrix} \frac{DV_g}{R(D-1)^2} \\ \frac{DV_g}{D-1} \end{bmatrix}. \quad (8)$$

Linearization with respect to duty gives

$$B_d = (A_{on} - A_{off})X + (B_{g,on} - B_{g,off})V_g = \begin{bmatrix} -\frac{V_g}{L(D-1)} \\ \frac{DV_g}{CR(D-1)^2} \end{bmatrix}. \quad (9)$$

Therefore

$$\dot{\hat{x}} = A\hat{x} + B_d\hat{d}, \quad \hat{v}_o = C_y\hat{x}, \quad (10)$$

and

$$G_{vd}(s) = \frac{\hat{v}_o(s)}{\hat{d}(s)} = C_y(sI - A)^{-1}B_d = \frac{V_g(-D^2R + DLS + 2DR - R)}{(D-1)^2(CLRs^2 + D^2R - 2DR + Ls + R)}. \quad (11)$$

Verification

- **PASS** — Rust MNA phase matrices equal the hand-derived phase equations.
- **PASS** — Rust numeric Schur reduction equals the SymPy descriptor reduction.
- **PASS** — Averaged A, input vector, DC point, and duty vector equal hand derivation.
- **PASS** — Input-to-output and control-to-output transfer functions simplify exactly.

All symbolic differences simplify exactly to zero. As a second check, the Rust numeric reducer and the SymPy descriptor reducer agree at $L = 100 \mu\text{H}$, $C = 220 \mu\text{F}$, and $R = 12 \Omega$.